

Shubham Chauhan

Linkedin: <https://www.linkedin.com/in/shubham-chauhan-spc/>

Email: shubham.spc90@gmail.com

GitHub: <https://github.com/Shubhamspc90>

Mobile: +91-7897754753

PROFESSIONAL SUMMARY

MCA graduate and aspiring AI/ML Engineer with strong expertise in Python, Machine Learning, SQL, and Generative AI. Experienced in developing end-to-end ML pipelines covering data preprocessing, feature engineering, model training, evaluation, and model persistence using Scikit-learn, Pandas, NumPy, and Matplotlib. Hands-on experience with LLM-based applications, LangChain, prompt engineering, structured output, chaining, and local LLMs using Ollama. Proficient in Git/GitHub with strong analytical, problem-solving, and software development skills. Passionate about building practical AI solutions, intelligent applications, and production-oriented Machine Learning and Generative AI systems.

SKILLS

-
- **Languages:** C, C++, Java, Python.
 - **Web Technologies:** HTML, CSS, JavaScript.
 - **Frameworks:** Flask, LangChain, Bootstrap.
 - **AI / Generative AI:** LLMs, Prompt Engineering, LangChain, Ollama, Structured Output
 - **Machine Learning:** Scikit-learn, Machine Learning, Data Preprocessing, Model Evaluation.
 - **Libraries/ GUI Tools:** Pandas, NumPy, Matplotlib, Tkinter, Java Swing, OpenCV.
 - **Data Preprocessing:** Pipeline, ColumnTransformer, OneHotEncoder, StandardScaler, Joblib.
 - **Tools/Platforms:** VS Code, IntelliJ IDEA, Git, GitHub.
 - **Database:** MySQL.
 - **Query Language:** SQL.
 - **Soft Skills:** Problem Solving, Time Management, Adaptability.

PROJECTS

Next Word Predictor | Python, TensorFlow, Keras, NLP, LSTM

Aug 2026 – Aug 2026

- Built an LSTM-based Next Word Prediction model using TensorFlow/Keras to predict the next word(s) in a sentence based on contextual patterns in text data.
- Designed and implemented an end-to-end NLP pipeline including text tokenization, sequence generation, padding, and label encoding to convert raw text into model-ready format.
- Engineered a deep learning architecture using an Embedding layer, LSTM layer, and Dense softmax output layer to effectively capture word sequence dependencies.
- Trained the model over multiple epochs to learn contextual word relationships, enabling accurate next-word predictions from user-provided input.
- Developed a real-time prediction feature that takes user input and iteratively generates the next N words, demonstrating practical use of sequence modeling in text generation.

Cat vs Dog Image Classification | TensorFlow, Keras, CNN, OpenCV, NumPy, Matplotlib

Aug 2026 – Aug 2026

- Built a CNN model for binary classification of cat and dog images using TensorFlow/Keras.
- Implemented image preprocessing including resizing, normalization, batching, and automatic label inference.
- Used Conv2D, MaxPooling, ReLU, Flatten, Dense, and Sigmoid layers for feature extraction and binary classification.
- Trained and evaluated the model using training and validation accuracy/loss curves to monitor model performance.
- Developed random test-image inference using OpenCV and compared actual vs predicted classes.

Uber Data Analysis | Python, Pandas, NumPy, Matplotlib, Seaborn

Aug 2026 – Aug 2026

- Performed Exploratory Data Analysis (EDA) on Uber ride request data to identify ride demand and trip status patterns.
- Cleaned and transformed timestamp data using Pandas and extracted request-hour features for time-based analysis.
- Analyzed ride requests by hour, pickup point, and trip status using filtering, grouping, aggregation, and frequency analysis.

- Developed data visualizations using Matplotlib and Seaborn to identify hourly ride trends and trip-status distributions.
- Conducted separate City and Airport pickup analysis and generated visualization outputs for comparative analysis.

Credit Card Customer Churn Prediction | Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib

July 2026 – July 2026

- Developed an Artificial Neural Network (ANN) using TensorFlow/Keras to predict customer churn.
- Performed data preprocessing, including One-Hot Encoding, feature scaling, and train-test split.
- Built and trained a feedforward neural network for binary classification.
- Evaluated model performance using Accuracy and Binary Cross-Entropy Loss.
- Saved the trained model and preprocessing pipeline for future inference.

Sales Profit Prediction Machine Learning Project | Python, Pandas, NumPy, Scikit-learn, Matplotlib

Jun 2026 – Jul 2026

- Designed and implemented an end-to-end Machine Learning workflow for sales profit prediction using Scikit-learn.
- Performed data preprocessing by handling missing values, encoding categorical features, scaling numerical data, and building reusable preprocessing pipelines using ColumnTransformer and Pipeline.
- Trained and compared six classification algorithms to select the optimal model based on multiple evaluation metrics.
- Evaluated model performance using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and Classification Report to identify the best-performing model.
- Saved the final trained model using Joblib and generated predictions on new customer data to demonstrate a complete end-to-end Machine Learning workflow.

Employees Management System | Java, Java Swing, JDBC, MySQL

Dec 2025 – Mar 2026

- Developed a desktop-based Employee Management System using Java Swing for the user interface and MySQL as the backend database.
- Established database connectivity using JDBC and performed CRUD operations on employee records.
- Implemented search and filtering functionality to enable efficient data retrieval.
- Displayed employee information using JTable for organized data presentation.
- Applied input validation and exception handling to ensure reliable system performance.
- Designed a user-friendly interface to improve usability and data management efficiency.

Face Recognition Attendance System | Python, Tkinter, OpenCV, MySQL, LBPH

July 2025 – Nov2025

- Developed a Face Recognition Attendance System using Python, OpenCV, Tkinter, and MySQL to automate attendance tracking.
- Implemented real-time face detection and recognition using the Local Binary Pattern Histogram (LBPH) algorithm.
- Designed and integrated a MySQL database for secure storage and management of student attendance records.
- Automated attendance marking and generated Excel-based attendance reports to reduce manual effort.
- Built a user-friendly Tkinter GUI dashboard for student registration, attendance monitoring, and record management.
- Utilized face recognition, image processing, and computer vision techniques to accurately identify individuals and improve attendance tracking efficiency.

College Leave Management System (CLMS) | PHP, MySQL, Bootstrap, PDO

Jan 2025 – May 2025

- Developed a full-stack College Leave Management System using PHP, MySQL, Bootstrap, and PDO to manage academic leave requests efficiently.
- Implemented role-based authentication and authorization for Admin, Teacher, and Student users with secure session management.
- Utilized PDO prepared statements and password hashing to ensure secure database interactions and prevent SQL injection attacks.
- Designed and managed a MySQL database for user records, leave applications, approval workflows, and status tracking.
- Developed leave application, approval, rejection, and tracking functionalities to automate and streamline leave

management processes.

- Built responsive administrative dashboards with search, filtering, and user management features using Bootstrap.

➤

E-Commerce Website | HTML, CSS, JavaScript, Bootstrap

Aug 2024 – Dec 2024

- Designed and developed a responsive E-Commerce website using HTML, CSS, JavaScript, and Bootstrap.
- Created dynamic product listing pages and implemented shopping cart functionality for an enhanced user experience.
- Developed a modern and intuitive user interface with responsive layouts using Bootstrap components.
- Optimized website performance and ensured cross-browser compatibility across multiple devices and platforms.
- Implemented mobile-first design principles to improve accessibility and responsiveness.
- Enhanced user navigation and product browsing experience through structured layouts and interactive UI elements.

CERTIFICATES

- | | |
|---|----------|
| ➤ Python Bootcamp: Learn Python from Scratch – Udemy's | Sep 2026 |
| ➤ Python with AI Workshop (30 Days) – AI For Techies. | Jul 2025 |
| ➤ Research Paper: "AI-Based Weather Prediction Models for Climate-Resilient Agriculture" – EAMIEEE Association. | Apr 2025 |
| ➤ Object-Oriented Programming Using C++ (150 Hours) – iamneo Platform. | Nov 2024 |
| ➤ One-Week Soft Skills Development Program. | Jan 2024 |
| ➤ Java Training Program (80 Hours) – SK Deft. | Jul 2022 |

EDUCATION

Lovely Professional University

Master of Computer Application (MCA): **CGPA: 7.43**

Punjab, India

May 2024 - Present

Dr. Virendra Swarup Institute of Computer Studies

Bachelor of computer Application (BCA): **CGPA: 8.0**

Kanpur Uttar Pradesh

Apr 2021 - May 2024

Shree Ram Convent School

Intermediate: **Percentage: 63%**

Varanasi Uttar Pradesh

Apr 2018 - March 2020

BPS Public School Naubatpur, Chandauli, Vns

High School: **Percentage: 71%**

Varanasi, Uttar Pradesh

Apr 2016 – Mar2018